

Detecting, quantifying, and attributing fundamental disagreements to annotator groups

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Abstract

Current annotation agreement metrics are often sensitive to group size imbalances and noise, making them inadequate for many subjective tasks such as toxicity and hate-speech detection. Polarization has recently been proposed as a more robust way to distinguish minor disagreements from systematic differences in opinion, but existing approaches do not provide practical tools for attributing polarization to specific annotator groups. We evaluate current methods and identify two major limitations in realistic settings: (1) the presence of “inherent” polarization that cannot be attributed to any known or latent groups, and (2) opposing polarization effects canceling each other out in aggregated annotations. To address these issues, we introduce a new metric that measures and tests the statistical significance of polarization attribution for annotator groups while avoiding these limitations. We apply our method to four subjective NLP datasets and find that gender and race consistently explain polarization patterns, while differences between annotator groups become stronger as the groups are further apart. Finally, we estimate the minimum number of annotators required for reliable results and release an open-source Python library implementing our approach.

1 Introduction

Annotations are essential for a wide range of tasks, especially in fields with subjective judgements, such as content moderation, toxicity, and hate speech detection. These tasks are especially important for training supervised learning models that remove or flag comments targeting vulnerable and minority groups in online spaces, where they are often vulnerable (United Nations, 2025; Dioum, 2025). However, annotations for such tasks are subjective and typically aggregated based on majority-voting (Ivey et al., 2025; Giannoutsos et al., 2025; Kralj Novak et al., 2022; van der Velden et al., 2025). This may at best limit the use-

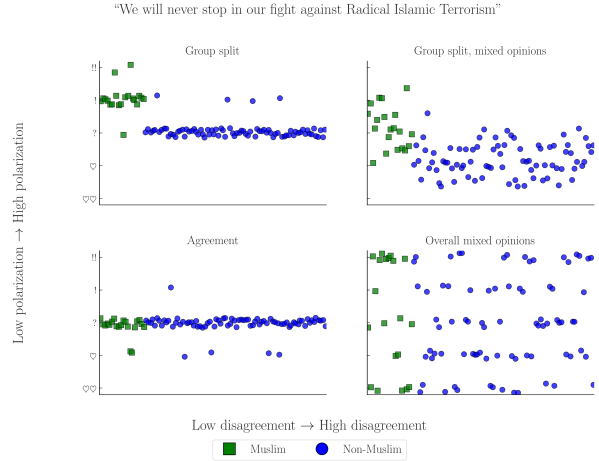


Figure 1: Simulated experiments for four scenarios on a hate speech detection task. Depending on the annotators, there may be a clear pattern where the split in annotations (polarization) can be attributed to a specific group (Muslim annotators in the case of the upper plots).

fulness of the data, or at worst lead to biased and misconfigured systems (Sap et al., 2019; van der Velden et al., 2025).

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree with the majority. These *fundamental* disagreements are typically ignored via aggregation (e.g., majority label voting) or filtering (e.g., keeping only items with large overall annotator agreement). For example, coded language may not be picked up by most annotators but only by ones belonging in the targeted minority (Quaranto, 2022), as shown in Fig. 1. Furthermore, majority-group annotators may themselves be biased against minority groups, e.g., toxicity models disproportionately marking African American (AA) speech as “toxic” (Sap et al., 2019).

Instead of using inter-annotator agreement, we could use *polarization*, which formulates annotation analysis as a cluster identification problem (Pavlopoulos and Likas, 2023, 2024). While we can detect polarization (Pavlopoulos and Likas, 2024), there is little work on how to detect whether

it is *actually caused by marginalized groups*. Our work thus tries to answer the following Research Question:

RQ. *Can we attribute polarization to annotator subgroups?*

We first identify limitations of current work on polarization:

1. Existing mechanisms (Pavlopoulos and Likas, 2024) operate only as binary classifiers, producing non-quantifiable/comparable results.
2. These mechanisms only test for the extreme case where *all* present polarization can be attributed to a single annotator group, which we find to be extremely rare in real-life datasets (App. B).
3. We are the first to report the existence of *inherent polarization*: most comments having some degree of polarization that cannot be attributed to *any possible partition* of annotators (Finding 1).
4. We also find that annotator groups may exhibit opposing polarization attribution between comments, requiring isolating annotations to each individual comment when analyzing whole datasets (Finding 2).

We then introduce a new metric (“Aposteriori Unimodality–*apunim*”) that attributes polarization to specific annotator groups without and develop a parametric test for statistical significance. Our metric is generalizable across any domain and modality (e.g., video/image classification) and, unlike previous approaches, enables *quantitative* comparisons across datasets. By considering the entire set of comments in a dataset *apunim* helps to identify genuine polarization and distinguish it from chance attribution, which is essential when analyzing large datasets.

Having established our metric, we apply it to four Natural Language Processing (NLP) datasets focused on Toxicity and Hate Speech detection (§5), finding that annotator gender and race significantly impacts annotation (Finding 3), and that fundamental disagreements grow more pronounced the further apart annotator groups are (e.g., irreligious vs. very religious annotators—Finding 4).

Finally, we release an open source Python library that implements our metric, which can be directly installed via PyPi.¹ and provide an estimation on the minimum number of annotators required for robust estimates (§6). The code for the experiments, graph creation, and analysis can be found on our project’s repository.²

¹Library: <https://pypi.org/project/apunim>
Documentation: <https://dimts-ts.github.io/apunim>

²Replication code: <https://github.com/dimts-ts/aposteriori-unimodality>

2 Related Work

2.1 Agreement

There are many popular metrics for measuring inter-annotator agreement such as Cohen’s kappa, Krippendorff’s alpha and Fleiss’s kappa. All of these metrics attempt to distinguish genuine disagreement among annotators from noise (or “random disagreement”) using statistical methods. However, most cannot be generally used for comparing intergroup annotation agreement without extending them (e.g., computing pairwise Cohen’s kappa for all groups), or placing constraints on when they can be used (e.g., no missing labels and same number of annotations per annotator for Fleiss’s kappa). They have also been criticized for ignoring fundamental differences that can arise from different social, cultural and demographic backgrounds (Feinstein and Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), for their inability to quantify and compare results across experiments (Wong et al., 2021), and for fundamentally faulty assumptions around the concept of “chance-correction” (Checco et al., 2017).

Recent work has attempted to work around these limitations by using simpler statistical models (van der Velden et al., 2025), adapting Cohen’s kappa for subgroup agreement (Wong et al., 2021), or creating a Bayesian model to explicitly differentiate between subgroup disagreement and noise (Ivey et al., 2025). While these extensions do improve the performance of these metrics, they are typically less explainable and intuitive.

2.2 Polarization

There are multiple competing formulations for polarization. One of these approaches (Akhtar et al., 2019) utilizes χ^2 to estimate in-group vs out-group variance, but still necessitates traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric (Checco et al., 2017), or non-parametric (Pavlopoulos and Likas, 2024; Mignemi et al., 2024) approaches to the annotation histograms.

For our work, we use a non-parametric polarization metric, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics (Pavlopoulos and Likas, 2024). nDFU takes values in the range $[0, 1]$, where values close to 0 indicate little or no polarization (i.e., opinions are concentrated around a single mode), while values close to 1 indicate strong polarization with multiple distinct peaks. The metric is defined as follows:

$$\text{nDFU}(X) = \frac{\max_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m} \quad (1)$$

$$m = \arg \max_i f_i$$

where, X denotes the set of annotations, f_i the relative frequency of the i -th annotation, and m the histogram peak. Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak.

2.3 Attributing polarization to groups

While many studies study the effect of subgroups on annotation agreement (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022), attributing polarization to subgroups is a relatively new idea. Under the name *a posteriori unimodality* (Pavlopoulos and Likas, 2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then those subgroups are “causing” the exhibited polarization. Though promising, this mechanism operates on an edge-case scenario where *all* polarization is explained by a single split. Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.³

3 What are the limitations of polarization attribution?

3.1 Problem Formulation

Let $d = \{c_1, c_2, \dots\}$ be a dataset composed of multiple annotated comments.⁴ Each comment c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \dots\}$, where a is a single annotation. As an example, the annotations for a comment in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\text{“Could be better, could be worse”}) \\ = \{ (+, F), (-, M), (+, F), \dots \}$$

Intuitively, a group partially explains the polarization of a comment c when the annotations of that group $A(c | g) = \{a(c; g') \in A(c) | g' = g\}$ exhibit higher polarization compared to the

³For example, when presented with noisy estimates of a metric in the $[0, 1]$ range, we would ideally want to distinguish between cases where attribution is close to 0.001, and cases where attribution is 0.49.

⁴These items could be of any modality, such as comments in a discussion, images, videos or survey questions, even though we consider comments in this study.

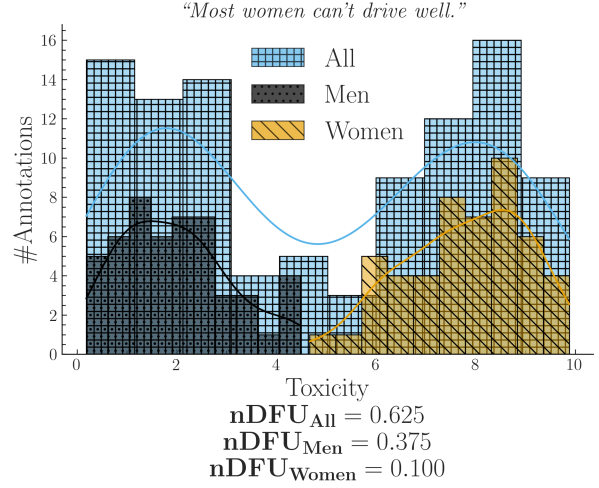


Figure 2: Synthetic experiment simulating a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

full set of annotations $A(c)$. Consider the (simulated) case where a misogynistic comment is annotated for toxicity by male and female annotators shown in Fig. 2. The annotations are generally polarized ($\text{nDFU}_{all} = 0.62$); but both female ($\text{nDFU}_{women} = 0.10$) and male ($\text{nDFU}_{men} = 0.37$) annotators exhibit low polarization between their respective groups. This pattern suggests that the overall polarization is driven by substantive disagreements between male and female annotators.

3.2 Inherent polarization

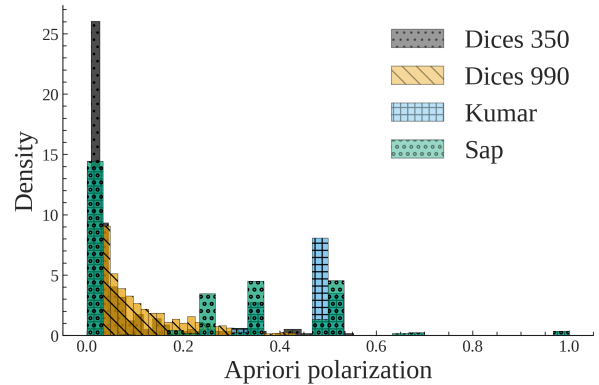


Figure 3: Apriori (inherent rather than observed) polarization across our datasets. Contrary to assumptions made in prior work, inherent polarization is rarely zero and can even reach $\text{nDFU} = 1$.

Inherent polarization refers to disagreement that cannot be attributed to *any specific* grouping of annotators; that is, polarization that exists no matter how we split the annotators in any arbitrary group. Prior work (Pavlopoulos and Likas, 2024)

attributed polarization to annotator groups only when the observed polarization for each individual group became exactly zero (e.g. $nDFU(all) > nDFU(men) = nDFU(women) = 0$). The presence of inherent polarization would invalidate this theory, since there would be no possible group g s.t. $nDFU(g) = 0$.

To examine this assumption, we create random annotator groupings for each comment in our datasets. Specifically, for every comment, we randomly partition the annotators into random groups (with randomly selected sizes) 100 times and compute the observed polarization for each partition. We then take the *minimum* observed polarization scores across all random partitions, which guarantees that we check for all possible variables, known and latent.

Formally, the inherent polarization for a single comment c is given by:

$$Pol_{inh}(c) = \min_i \{nDFU(\tilde{r}_i(A(c)))\} \quad (2)$$

where $\tilde{r}$ is the random partition operator.

Repeating this process for every comment produces the distribution shown in Fig. 3, demonstrating that inherent polarization is consistently non-zero and can even be substantial for many comments.

Finding 1. *Each comment may exhibit a degree of inherent polarization that cannot be attributed to any known or latent annotator grouping.*

3.3 Conflicting polarization

Previous work (Pavlopoulos and Likas, 2024) operated on the comment level. In practice however, this approach is inherently noisy due to a lack of information (i.e., limited annotations) relating to each individual comment. Many datasets in NLP are annotated by only 3–5 annotators per comment, which is especially problematic when we want to analyze patterns in annotator subgroups; in a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split. This issue is especially prevalent for polarization, since $nDFU$ is undefined for groups of two (and arguably marginally useful for groups of three), since it requires at least two histogram peaks to function (see Eq. 1), a limitation not acknowledged in the original work.

This limitation leaves us with two choices: (1) only use datasets featuring at least 6–8 annotators, or (2) aggregate annotations across comments for each dataset. The former case is unrealistic, especially for most NLP datasets due to annotation costs. We consider the latter case in an experiment shown in Fig. 4 where we combine annotations across two comments. We observe that, should the two comments be polarized for the opposite reason, the opposing polarization effects might bal-

ance each other out, leading to a false negative. Therefore, we should never aggregate annotations across comments.⁵

Finding 2. *Polarization has a direction, which may compromise cross-comment annotation aggregation.*

4 How do we fix it?

4.1 From inherent to apriori polarization

In §3.1, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c | g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 2, we would end up subtracting the $nDFU$ s of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

We instead compare the observed polarization of $A(c)$ with its inherent polarization, following the methodology introduced in §3.2 with two key distinctions. (1) Instead of using random splits, we create random stratified splits.⁶ (2) Instead of finding the minimum possible polarization exhibited by any known or latent group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups*, which is a much stricter test.⁷ Specifically, we define a-priori unimodality for a comment c as:

$$Pol_{apr}(c) = \frac{1}{t} \sum_{i=1}^t nDFU(\tilde{r}_i(A(c))) \quad (3)$$

4.2 Expanding to the whole dataset

As discussed in §3.3, we must not aggregate the annotations themselves before applying the $nDFU$ metric. We can however estimate how much polarization is generally present in the dataset, by simply averaging the *a-priori* polarization of its comments. Intuitively, a high apriori polarization

⁵A similar assumption was followed in early parametric approaches, which assumed different parameters for each comment (Checco et al., 2017).

⁶I.e., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we create random partitions of sizes 80 and 20.

⁷Indeed, by definition, all groups exhibit equal or larger polarization than the inherent pol. value.

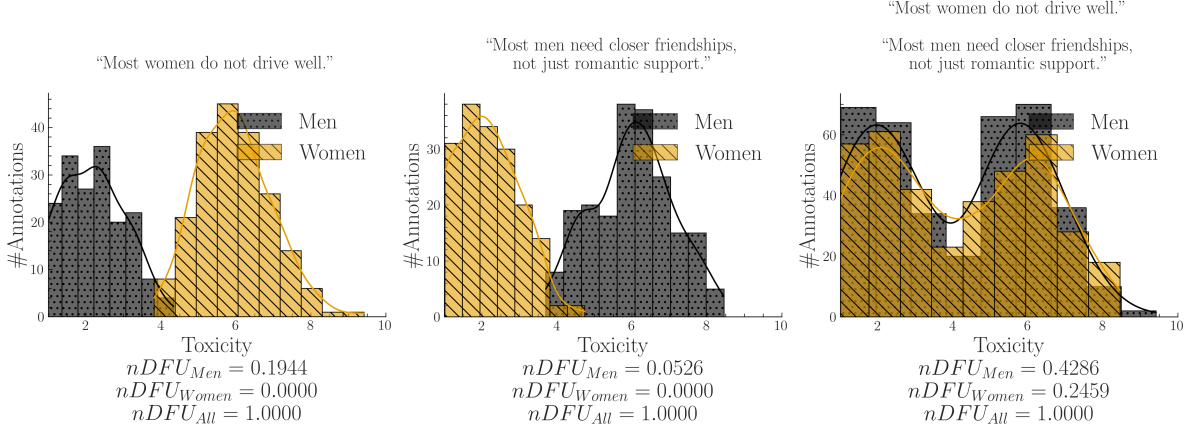


Figure 4: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization.

score means that much of the polarization in the dataset is likely not driven by any group.

$$Pol_{apr}^d = \frac{1}{|d|} \sum_{c \in d} Pol_{apr}(c) \quad (4)$$

Which we will compare with the average observed polarization of all comments:

$$Pol_{obs}^d(g) = \frac{1}{|d|} \sum_{c \in d} nDFU(A(c | g)) \quad (5)$$

We can now compare the polarization that is attributed to a single-dimensional annotator group (Equation 5) to a prior one (Equation 4). We define our normalized metric, which we call “*apunim*” (*Aposteriori Unimodality*), as:

$$apunim(d, g) = \frac{Pol_{obs}^d(g) - Pol_{apr}^d}{1 - Pol_{apr}^d} \quad (6)$$

Since $nDFU \rightarrow [0, 1] \Rightarrow apunim \rightarrow [-1, 1]$. The interpretation of this metric boils down to three scenarios:

- $apunim(\theta) \approx 0$: The examined group does not meaningfully disagree with the other annotators.
- $apunim(\theta) > 0$: The examined group has fundamental disagreements with the other annotators.
- $apunim(\theta) < 0$: Overall polarization *drops* when we add annotators from this group to this task. This occurs when intragroup polarization is higher than apriori polarization; indicating that annotators of that group are polarized between themselves.

4.3 Statistical significance

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized partitions and compare the observed apunim against this distribution using a one-sample Student-*t* test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$H_0 : apunim(d, g) \sim apunim_{rand}(d, g)$$

$$H_a : apunim(d, g) \not\sim apunim_{rand}(d, g)$$

where $apunim_{rand}(d, g)$ denotes the distribution of apunim values obtained from randomized annotator partitions. For a detailed explanation of the p-value computation and the assumptions underlying the test, refer to App. C.

5 Attributing real-life polarization to human groups

5.1 Datasets

We use four datasets from current NLP research that include sociodemographic information at the level of individual annotators; specifically, the “Kumar” (Kumar et al., 2021) and “Sap” (Sap et al., 2022) datasets, which are concerned with Toxicity and hate speech detection respectively, and the DICES-350 and DICES-990 datasets (Aroyo et al., 2023), which focus on LLM response safety.⁸ While the toxicity and hate

⁸For the latter, we consider only labels relating to racism (Q3_bias_overall), thereby transforming the task to hate speech detection.

Dataset	#Com	#Ann	#Dims	Task
Kumar	106,035	5	10	Tox
Sap	626	5–6	3	HS
DICES-350	72,103	70–76	4	HS
DICES-990	43,050	123	4	HS

Table 1: Datasets used in this study. $\#Ann$ refers to the number of annotations per comment (see App. B), and $\#Dims$ to sociodemographic dimensions (e.g., age, gender). Tasks are either toxicity (Tox) or hate speech detection (HS).

speech detection tasks are not equivalent, they share substantial conceptual and operational overlap, as both require annotators to assess the presence and severity of harmful or abusive language. Thus, we expect the sources of annotator disagreement and bias that arise in these tasks will be broadly similar.

A brief overview of dataset characteristics is provided in Table 1. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct sociodemographic dimensions. In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. The Kumar and Sap datasets contain a large number of polarizing items, unlike the DICES datasets, where most items are unpolarized (Fig. 7).⁹

5.2 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided sociodemographic dimensions can partially explain polarization in the toxicity / hate speech annotations, as shown in Table 1.¹⁰ Note that some groups are not included, either because the results were not statistically significant, or because there were no polarized items annotated by more than one annotator of these groups. See App. E for details on other parameters and execution times, and App. F for the full results.

5.3 Which factors contribute to polarization?

Race Across all datasets apart from Sap, we observe a consistent pattern in which annotation polarization is linked to annotator characteristics, particularly ethnicity. In the hate speech detection datasets (DICES-350, DICES-990) *ethnicity is the*

Group	apunim	support
DICES-350		
Race=AfricanAmerican	-0.043	9077
Race=Asian	0.066	8138
DICES-990		
Age=GenX+	-0.046	14384
Age=GenZ	0.063	13741
Edu=High school and below	-0.075	7419
Gender=Man	0.028	32494
Gender=Woman	-0.025	33471
Race=AfricanAmerican	0.091	5810
Race=Latino	-0.133	6610
Race=Other	0.075	24321
Race=White	-0.120	11392
Sap		
Gender=Man	0.091	1439
Gender=Woman	-0.244	877
Kumar		
Age=18–24	0.611	331
Age=25–34	0.606	8506
Age=35–44	0.557	2556
Age=45–54	0.555	441
Age=55–64	0.637	110
Edu=Masters	0.642	858
Edu=Bachelors	0.594	9013
Edu=Associate degree	0.585	335
Edu=College, no degree	0.570	1425
Edu=High School graduate	0.446	144
Eth=Asian	0.461	49
Eth=Black	0.618	468
Eth=White	0.546	26239
Gender=Female	0.586	14479
Gender=Male	0.588	11716
PastTargeted=No	0.551	25773
PastTargeted=Yes	0.608	3893
Parent=No	0.566	10784
Parent=Yes	0.593	15929
Trans=No	0.484	40404
Politics=Conservative	0.620	3075
Politics=Independent	0.574	3270
Politics=Liberal	0.580	7897
ReligionImportant=No	0.555	4041
ReligionImportant=Low	0.572	315
ReligionImportant=Mid	0.624	2542
ReligionImportant=High	0.631	5403
SeenToxicity=No	0.627	2454
SeenToxicity=Yes	0.529	28821
SexOrientation=Bi	0.621	467
SexOrientation=Het	0.528	32217
TechImportant=Negative	0.492	217
TechImportant=Neutral	0.587	426
TechImportant=Positive	0.579	13685
TechImportant=Very pos.	0.622	2942
ToxProblem=Never	0.451	55
ToxProblem=Rare	0.572	1007
ToxProblem=Occasionally	0.588	6082
ToxProblem=Freq	0.579	3961
ToxProblem=VeryFreq	0.588	410

Table 2: Statistically significant ($p < 0.05$) a posteriori unimodality results across all four datasets. Gray rows show negative values. Consult App. F for the full results.

⁹This systematic discrepancy could be attributed to the DICES datasets being composed of Large Language Model (LLM)-generated dialogues with already-aligned models tuned to avoid racist speech.

¹⁰We sample 10,000 comments from the Kumar dataset due to computational costs arising from the extraordinary amount of Personal Characteristics (PCs).

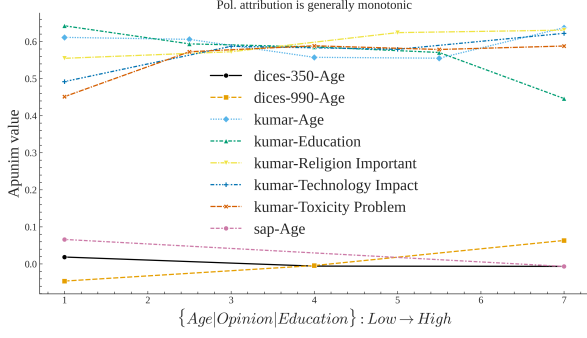


Figure 5: Polarization attribution trends in ordinal groups. The left side of the x-axis (low orders) refers to young annotators, low education, low religiousness, and indifference towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in App. F.

strongest contributor to polarization, with statistically significant effects for at least one major ethnic group in each dataset (AAs: -0.0428 /Asians: 0.0659 for DICES-350, Whites: -0.0377 for Kumar, and all ethnicities for DICES-990). These results align with prior work suggesting that ethnic background is a systematic source of annotator disagreement rather than isolated noise (Goyal et al., 2022), validating our initial motivation.

Gender Polarization is also consistently linked to gender across all datasets except DICES-350. In DICES-990, men and women show statistically significant effects in opposite directions (Men: 0.028 , Women: -0.025), mirrored in Sap, where men have a positive attribution score (0.091) and women a strongly negative one (-0.244). By comparison, the Kumar dataset shows almost identical attribution scores for female and male annotators (Female: 0.586 , Male: 0.588), a pattern that also holds for many other attributes in that dataset. This may indicate that *apunim* is partly task-dependent, or that the test becomes less discriminative when many comments are annotated by relatively few annotators. Nevertheless, these patterns mirror prior findings with respect to hate speech (Wojatzki et al., 2018) and toxicity (Binns et al., 2017) annotation.

Finding 3. *Annotator age and race are consistent causes of polarization.*

Polarization attribution follows a monotonic trend Figure 5 shows *apunim* trends across ordinal sociodemographic attributes listed in Table 2. We find that polarization attribution changes consistently in one direction as we move from one end of an attribute scale to the other—for example, from irreligious to very religious annotators. This monotonic pattern appears across *all* ordinal sociode-

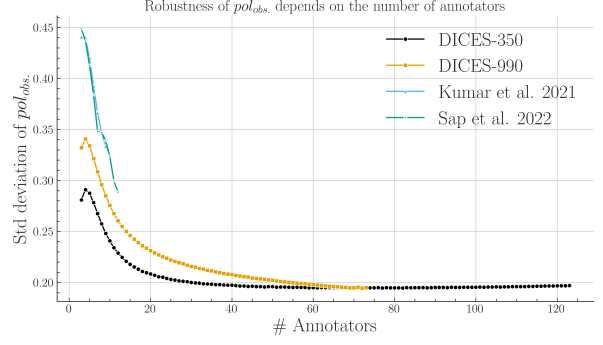


Figure 6: Standard deviation of 30 randomly sampled (with replacement) pol_{obs} values (Eq. 5). The x-axis shows the sample size (number of annotators), starting from 3 and increasing one at a time, up to the largest annotator count for which a sufficient number of items exists (B).

mographic dimensions and *all* datasets (apart from Age in the Kumar dataset, which is U-shaped), suggesting that perceived differences become stronger between groups that are further apart ideologically or demographically.

Finding 4. *For ordinal sociodemographic dimensions, polarization attribution tends to grow stronger as the differences between annotator groups increase.*

6 How many annotators do we need?

While *apunim!* (*apunim!*) works best with many annotations per item, the cost required for multiple human annotators is often greatly restrictive (Rossi et al., 2024). Thus, it is imperative to find the least annotators needed for *apunim* to generate stable and reliable estimates. For each item per dataset, we randomly sample with replacement a subset of annotators and compute the standard deviation of 30 sampled pol_{obs} values (see Equation 5). Figure 6 demonstrates the effect of the number of annotators on the reliability of the polarization estimation, which we discuss next.

Each dataset features a different standard deviation even with the same number of annotators, which is broadly consistent with the amount of polarization in each dataset (App. B). However, we observe a clear, non-linear, inverse relationship between the number of annotators and the standard error across datasets. This can be attributed to a more robust histogram estimation used by *nDFU* (§4.1) when provided with more points for each of the subgroups. In the case of our datasets, we observe that after about 20 points, the estimation is good enough that more annotations contribute trivially to the robustness of the metric.

7 Conclusion & Future work

We investigated whether fundamental differences can be attributed to annotator groups using polarization, instead of standard agreement measures. We discovered as of yet unknown issues regarding the presence of inherent polarization and conflicting polarization directions which we bypass using a new metric, **apunim!**. We then applied our proposed metric to four hate-speech and toxicity detection datasets, discovering that (1) race and gender seem to generally drive polarization attribution, and (2) groups that are further apart increasingly drive polarization. Furthermore, we estimated the robustness of our metric w.r.t. the number of annotators, and released a PyPi installable library implementing it.

Our framework has broad applications across NLP tasks related to content moderation. Because many of these tasks involve multiple discussions, the framework can also be applied at the discussion level rather than only at the dataset level. This would allow researchers to identify particularly polarizing discussions and study the underlying sources of disagreement for specific minority groups. Discussions could also be analyzed by comment turn, enabling the study of how polarization develops over the course of an interaction. Finally, our approach can be used to more reliably evaluate whether LLM-as-a-judge systems reproduce the polarization patterns observed among different minority groups.

8 Limitations

Researchers often release only aggregated single-label annotations or omit annotator sociodemographic characteristics entirely. As a result, both our analysis and the applicability of our framework are limited by the scarcity of suitable datasets. In addition, because datasets with large numbers of annotators per item are rare, our approach operates only at the dataset level. This limitation prevents us from identifying individual comments or examples that could serve as concrete explanations for the observed *apunim* values. Finally, although we compare findings across datasets and relate them to prior work, we do not directly compare polarization and agreement metrics, since they rely on fundamentally different assumptions and mechanisms (see §2.2). This also makes it difficult to fully explain certain patterns in our results, such as why some groups show statistically significant effects while others do not.

9 Ethical Considerations

All datasets used in this study were either publicly accessible, or were provided to us by the authors with full disclosure on the purpose of our research.

No identifiable, personal information is included in our pipeline. We did not perform any annotation tasks ourselves.

References

- Sohail Akhtar, Valerio Basile, and Viviana Patti. 2019. A new measure of polarization in the annotation of hate speech. In *AI*IA 2019 – Advances in Artificial Intelligence*, pages 588–603, Cham. Springer International Publishing.
- Hala Al Kuwatly, Maximilian Wich, and Georg Groh. 2020. Identifying and measuring annotator bias based on annotators’ demographic characteristics. In *Proceedings of the fourth workshop on online abuse and harms*, pages 184–190.
- Lora Aroyo, Alex Taylor, Mark Díaz, Christopher Homan, Alicia Parrish, Gregory Serapio-García, Vinodkumar Prabhakaran, and Ding Wang. 2023. *Dices dataset: Diversity in conversational ai evaluation for safety*. In *Advances in Neural Information Processing Systems*, volume 36, pages 53330–53342. Curran Associates, Inc.
- Reuben Binns, Michael Veale, Max Van Kleek, and Nigel Shadbolt. 2017. Like trainer, like bot? inheritance of bias in algorithmic content moderation. In *International conference on social informatics*, pages 405–415. Springer.
- Ted Byrt, Janet Bishop, and John B. Carlin. 1993. *Bias, prevalence and kappa*. *Journal of Clinical Epidemiology*, 46(5):423–429.
- Alessandro Checco, A. Roitero, Eddy Maddalena, S. Mizzaro, and Gianluca Demartini. 2017. Let’s agree to disagree: Fixing agreement measures for crowdsourcing.
- S.Y. Chen, Z. Feng, and X. Yi. 2017. A general introduction to adjustment for multiple comparisons. *Journal of Thoracic Disease*, 9(6):1725–1729. Doi: 10.21037/jtd.2017.05.34; PMID: 28740688; PMCID: PMC5506159.
- Shelby Dioum. 2025. *What explains the increase in online hate speech?* Accessed: 2025-09-22.
- Alvan R. Feinstein and Domenic V. Cicchetti. 1990. *High agreement but low kappa: I. the problems of two paradoxes*. *Journal of Clinical Epidemiology*, 43(6):543–549.
- Sokratis-Vissarion Giannoutsos, Dimitris Tsirmpas, and John Pavlopoulos. 2025. Llm depolarization via aposteriori unimodality.
- Tommaso Giorgi, Lorenzo Cima, Tiziano Fagni, Marco Avvenuti, and Stefano Cresci. 2025. Human and llm biases in hate speech annotations: A socio-demographic analysis of annotators and targets. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 19, pages 653–670.

617	Nitesh Goyal, Ian D Kivlichan, Rachel Rosen, and	theory: A case study in a hate speech corpus.	674
618	Lucy Vasserman. 2022. Is your toxicity my tox-	In <i>Proceedings of the 2022 ACM conference on</i>	675
619	icity? exploring the impact of rater identity on	<i>fairness, accountability, and transparency</i> , pages	676
620	toxicity annotation. <i>Proceedings of the ACM on</i>	1585–1603.	677
621	<i>Human-Computer Interaction</i> , 6(CSCW2):1–28.		
622	Sture Holm. 1979. A simple sequentially rejective	Maarten Sap, Dallas Card, Saadia Gabriel, Yejin	678
623	multiple test procedure. <i>Scandinavian Journal</i>	Choi, and Noah A. Smith. 2019. The risk of	679
624	<i>of Statistics</i> , 6(2):65–70.	racial bias in hate speech detection. In <i>Proceed-</i>	680
625	Jonathan Ivey, Susan Gauch, and David Jurgens.	<i>ings of the 57th Annual Meeting of the Associa-</i>	681
626	2025. NUTMEG: Separating signal from noise	<i>tion for Computational Linguistics</i> , pages 1668–	682
627	in annotator disagreement. In <i>Proceedings of</i>	1678, Florence, Italy. Association for Computa-	683
628	<i>the 2025 Conference on Empirical Methods in</i>	tional Linguistics.	684
629	<i>Natural Language Processing</i> , pages 2874–2887,		
630	Suzhou, China. Association for Computational	Maarten Sap, Swabha Swayamdipta, Laura	685
631	Linguistics.	Vianna, Xuhui Zhou, Yejin Choi, and Noah A.	686
632	Petra Kralj Novak, Teresa Scantamburlo, Andraž	Smith. 2022. Annotators with attitudes: How	687
633	Pelicon, Matteo Cinelli, Igor Mozetič, and Fabi-	annotator beliefs and identities bias toxic lan-	688
634	ana Zollo. 2022. Handling disagreement in hate	guage detection. In <i>Proceedings of the 2022 Con-</i>	689
635	speech modelling. In <i>Information Processing</i>	<i>ference of the North American Chapter of the</i>	690
636	<i>and Management of Uncertainty in Knowledge-</i>	<i>Association for Computational Linguistics: Hu-</i>	691
637	<i>Based Systems</i> , pages 681–695, Cham. Springer	<i>man Language Technologies</i> , pages 5884–5906,	692
638	International Publishing.	Seattle, United States. Association for Computa-	693
639	Deepak Kumar, Patrick Gage Kelley, Sunny Con-	tational Linguistics.	694
640	solvo, Joshua Mason, Elie Bursztein, Zakir Du-		
641	rumeric, Kurt Thomas, and Michael Bailey.	Ole Tange. 2025. Gnu parallel 20250822. GNU	695
642	2021. Designing toxic content classification for	Parallel is a general parallelizer to run multiple	696
643	a diversity of perspectives. In <i>Proceedings of the</i>	serial command line programs in parallel with-	697
644	<i>Seventeenth USENIX Conference on Usable Pri-</i>	out changing them.	698
645	<i>vacancy and Security</i> , SOUPS’21, USA. USENIX		
646	Association.	United Nations. 2025. Targets of hate. Accessed:	699
647	Giuseppe Mignemi, Antonio Calcagnì, Andrea	2025-09-22.	700
648	Spoto, and Ioanna Manolopoulou. 2024. Mix-	Mariken A. C. G. van der Velden, Felicia	701
649	ture polarization in inter-rater agreement analy-	Loeberbach, Wouter van Atteveldt, Antske	702
650	sis: a bayesian nonparametric index. <i>Statistical</i>	Fokkens, Myrthe Reuver, and Kasper Welbers.	703
651	<i>Methods & Applications</i> , 33(1):325–355.	2025. Whose truth is it anyway? an experiment	704
652	John Pavlopoulos and Aristidis Likas. 2023. Dis-	on annotation bias in times of factual opinion	705
653	tance from unimodality for the assessment of	polarization. <i>Communication Methods and Mea-</i>	706
654	opinion polarization. <i>Cognitive Computation</i> ,	<i>sures</i> , 19(4):332–349.	707
655	15(2):731–738.		
656	John Pavlopoulos and Aristidis Likas. 2024. Po-	Michael Wojatzki, Tobias Horsmann, Darina Gold,	708
657	larized opinion detection improves the detection	and Torsten Zesch. 2018. Do women perceive	709
658	of toxic language. In <i>Proceedings of the 18th</i>	hate differently: Examining the relationship be-	710
659	<i>Conference of the European Chapter of the As-</i>	tween hate speech, gender, and agreement judg-	711
660	<i>sociation for Computational Linguistics (Volume</i>	ments.	712
661	<i>1: Long Papers)</i> , pages 1946–1958, St. Julian’s,		
662	Malta. Association for Computational Linguis-	Ka Wong, Praveen Paritosh, and Lora Aroyo. 2021.	713
663	tics.	Cross-replication reliability - an empirical ap-	714
664	Anne Quaranto. 2022. Dog whistles, covertly	proach to interpreting inter-rater reliability. In	715
665	coded speech, and the practices that enable	<i>Proceedings of the 59th Annual Meeting of the</i>	716
666	them. <i>Synthese</i> , 200(4):330.	<i>Association for Computational Linguistics and</i>	717
667	Luca Rossi, Katherine Harrison, and Irina	<i>the 11th International Joint Conference on Nat-</i>	718
668	Shklovski. 2024. The problems of llm-generated	<i>ural Language Processing (Volume 1: Long Pa-</i>	719
669	data in social science research. <i>Sociologica</i> ,	<i>pers)</i> , pages 7053–7065, Online. Association for	720
670	18(2):145–168.	Computational Linguistics.	721
671	Pratik S Sachdeva, Renata Barreto, Claudia von	Appendix A Acronyms	722
672	Vacano, and Chris J Kennedy. 2022. Assessing	LLM Large Language Model	723
673	annotator identity sensitivity via item response	PC Personal Characteristic	724
		nDFU normalized Distance From Unimodality	725
		FWER Family-Wise Error Rate	726
		NLP Natural Language Processing	727

Appendix B Annotation details

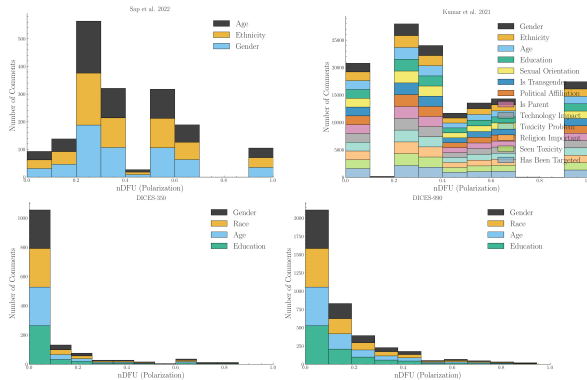


Figure 7: Histogram of dataset polarization for the human datasets considered in this paper per sociodemographic.

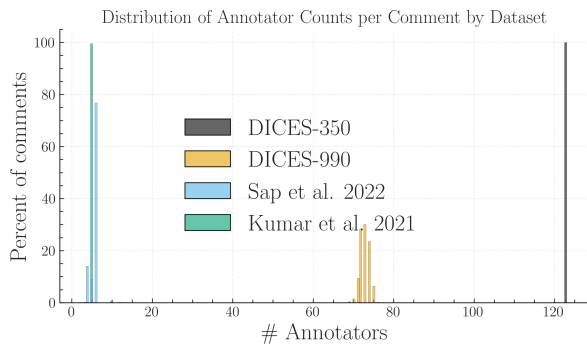


Figure 8: We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table 3).

Table 3: Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73	74	76
Kumar et al. 2021	20000	5.0925	4.8872	5	5	5	5	12
Sap et al. 2022	585	5.6359	0.7713	3	5	6	6	12

The number of annotators per comment for all datasets presented in §?? can be found in Figure 8, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used

from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We included these comments for the experiment presented in Table 2, but not for the annotator number experiment in Figure 6 since they would produce uninterpretable figures.

Appendix C P-value estimation implementation

Since our test is parametric, it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per sociodemographic group to estimate $nDFUs$ in Equations 3-5, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §??). We discuss condition (2) in §6. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single θ dimension in principle, in practice we often want to test polarization for all subgroups ($\forall \theta \in \Theta$). Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method (Holm, 1979). The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses (Chen et al., 2017)). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

Appendix D Software

D.1 Installation and usage

We release the apunim package, which is a Python library that implements both the $nDFU$ and apunim calculations, as well as its statistical test.

The software is available on PyPi (<https://pypi.org/project/apunim>). We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (<https://dimits-ts.github.io/apunim>).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and

can be found on the project’s repository (<https://github.com/dimits-ts/apunim>). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy`, `scipy`, `statsmodels`).

D.2 Implementation details

By default, the computation of the `apunim` values happens on a single thread. In cases of serious computational cost, we recommend running the `apunim` calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple sociodemographic dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{P}_i$, which are originally calculated to obtain the apriori polarization values (see Equation 3), for the p-value calculations (§??). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{P}_i$ in Equation 3), and the number of `apunim` values used in the p-value calculation ($\tilde{D}_i$ in Algorithm ??) have to be the same.

Appendix E Replication Details

We use version 1.0.1 of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Equation 3) and the p-value estimation (§??), and set a `FWER` rate of 0.95.

The current version of the library handles large number of annotators very well due to the use of vectorized operations, but struggles when faced with a lot of comments when the annotators belong to multiple groups. Indeed, while the experiments for all datasets presented in §?? need no more than a minute to run, the Kumar dataset requires about 12 hours of computation even with a 20% sample. The annotator variance analysis presented in §6 also required more than 10 hours of computation, although that is expected due to the nature of the experiment, which is very computationally demanding, and a non-standard use of our library.

This bottleneck is likely caused by repeated creation of `numpy` arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each sociodemographic dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each sociodemographic subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` linux package (Tange, 2025) for our own experiments).

In Figure 6, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-

990 were annotated by more than 73 annotators—see B).

Appendix F Full Results

Table 4: Aposteriori unimodality results for the DICES-350 dataset.

sociodemographic dimension	Value	apunim
Age	1) Gen. X+	0.0186
	2) Millenial	-0.0059
	3) Gen. Z	-0.0066
Education	College or higher	0.0124
	High school or lower	-0.0152
	Other	-0.0214*
Gender	Man	-0.0193
	Woman	0.0215
Race	African Am.	-0.0428
	Asian	0.0659
	Latino	-0.0030
	Multiracial	-0.0001
	White	-0.0198

Table 5: Aposteriori unimodality results for the DICES-990 dataset.

sociodemographic dimension	Value	apunim
Age	1) Gen. X+	-0.0471
	2) Millenial	-0.0043
	3) Gen. Z	0.0644
Education	College or higher	0.0095
	High school or lower	-0.0751
Gender	Man	0.0278
	Woman	-0.0235
Race	African Am.	0.0909
	Latino	-0.0030
	Multiracial	-0.0001
	White	-0.0198

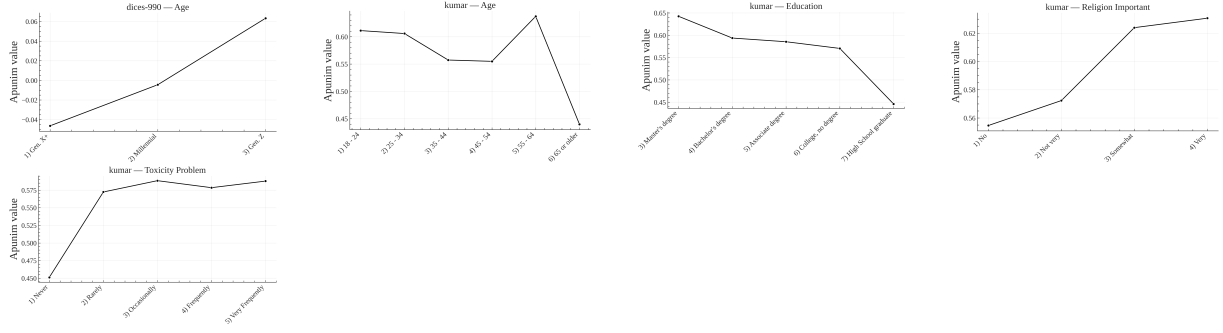


Figure 9: Apunim values across ordinal factor levels with at least two statistically significant values, expanded for each ordinal sociodemographic. This figure is complementary to Figure 5. The values in the graphs were extracted from the Tables of F.

Table 6: Aposteriori unimodality results for the Sap dataset.

sociodemographic dimension	Value	apunim	support
Age	1) Gen. X+	0.0192	743
	2) Millennial	0.0189	1792
	3) Gen. Z	-0.0153	239
Ethnicity	Black	-0.2967	1041
	White	0.1400	2048
Gender	Male	0.0942	1635
	Female	-0.1810	1333